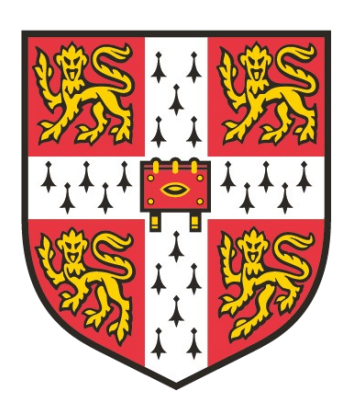


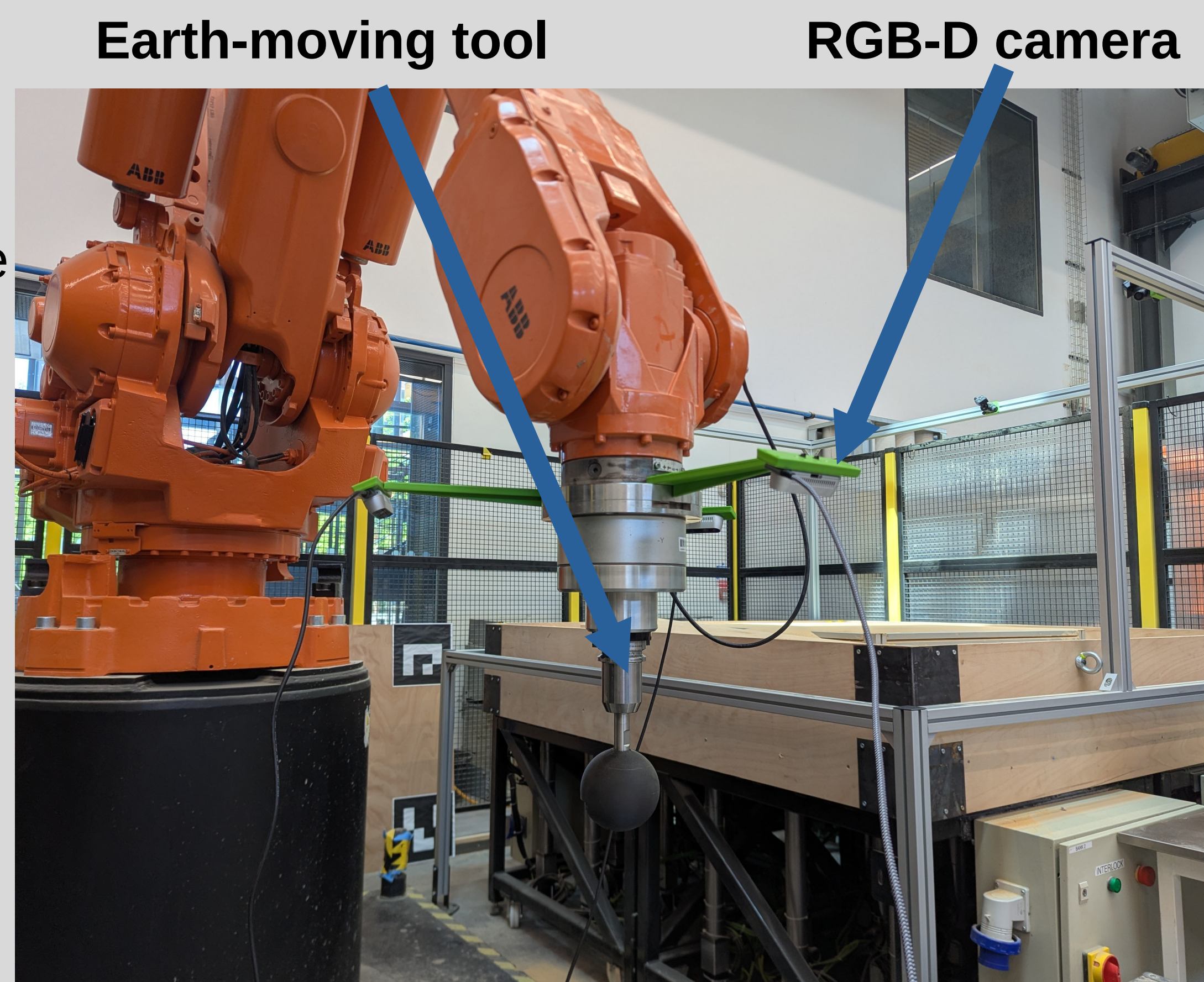


J.Ingham – come find me!

Eye-in-hand computer vision for autonomous geotechnical assessment – MaGIC 2026

Motivation

- Autonomous earth moving requires knowledge of earth state (visually)
- Lack of earth moving datasets
- Can cheap RGBD cameras replace expensive industry-standard LiDAR?
- Can point clouds be compressed and still retain geotechnical information?



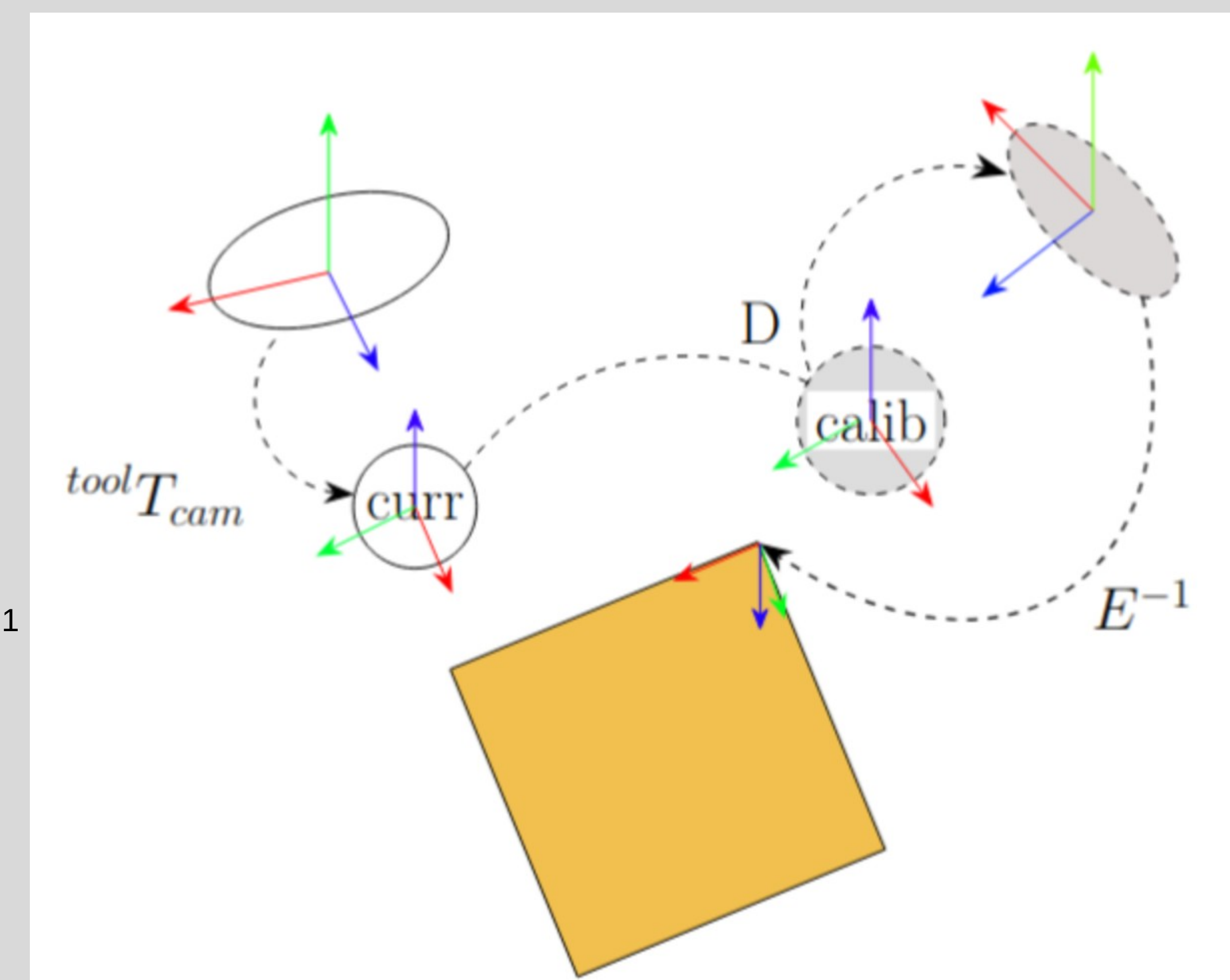
Terra Robotics Laboratory robot and multi-RGBD setup

Transformation

- RGB-D cameras capture point clouds (p_{sense}) that need to be transformed into the workspace (p_{task})
- By calibrating with a ChArUco board, the camera extrinsics E and ${}^{\text{tool}}T_{\text{cam}}$ can be calculated

$$p_{\text{task}} = E^{-1}D{}^{\text{tool}}T_{\text{cam}}p_{\text{sense}}$$

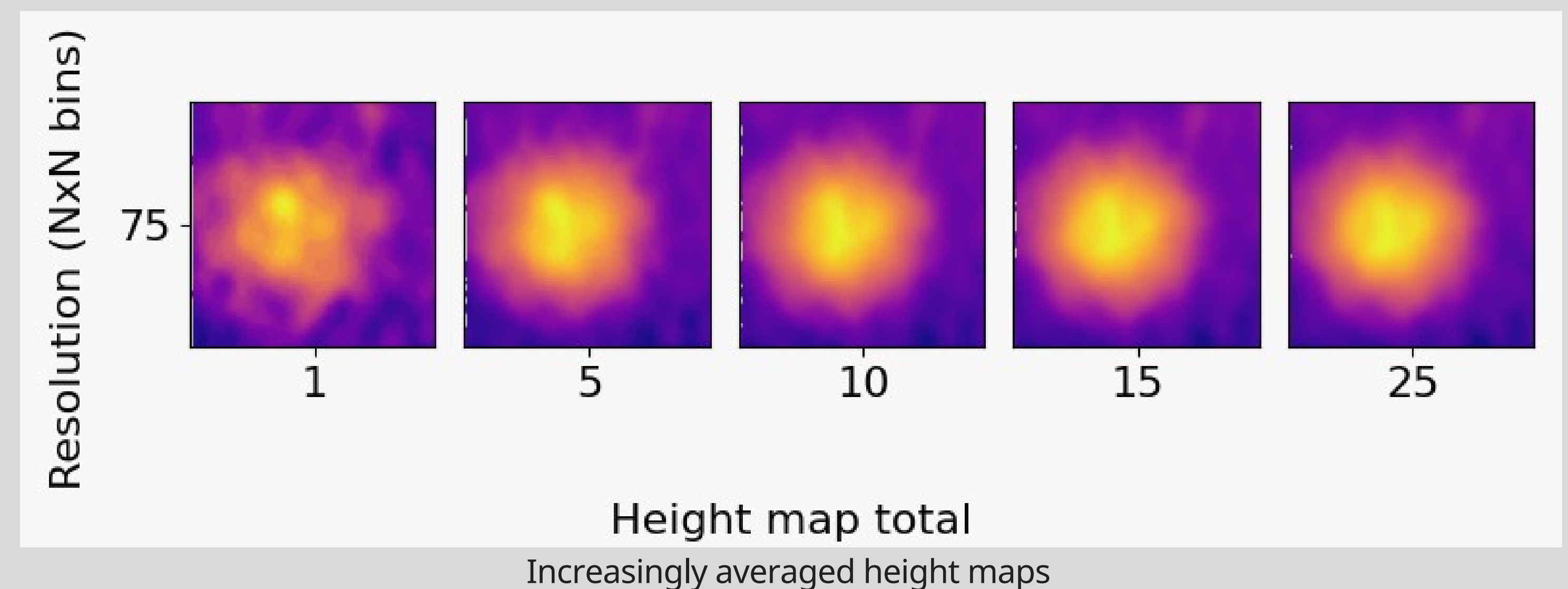
- D is the transformation from the current camera position to the calibration position



Point cloud transformation

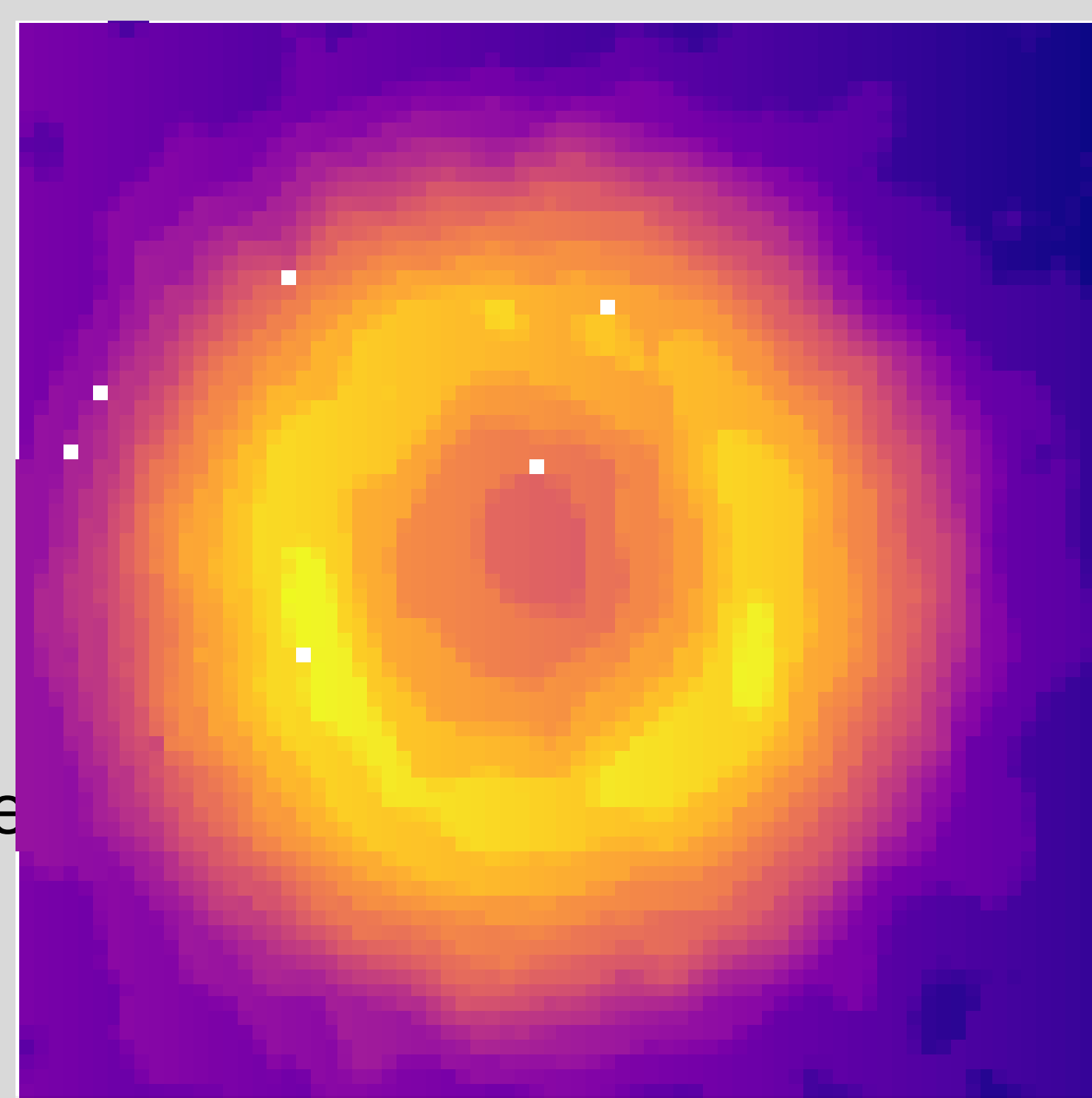
Solo camera

- Single camera directly above the feature
- Compared at various distances with industry standard LiDAR scanner
- At further distances, time-based heightmap averaging is required to improve accuracy



Heightmaps

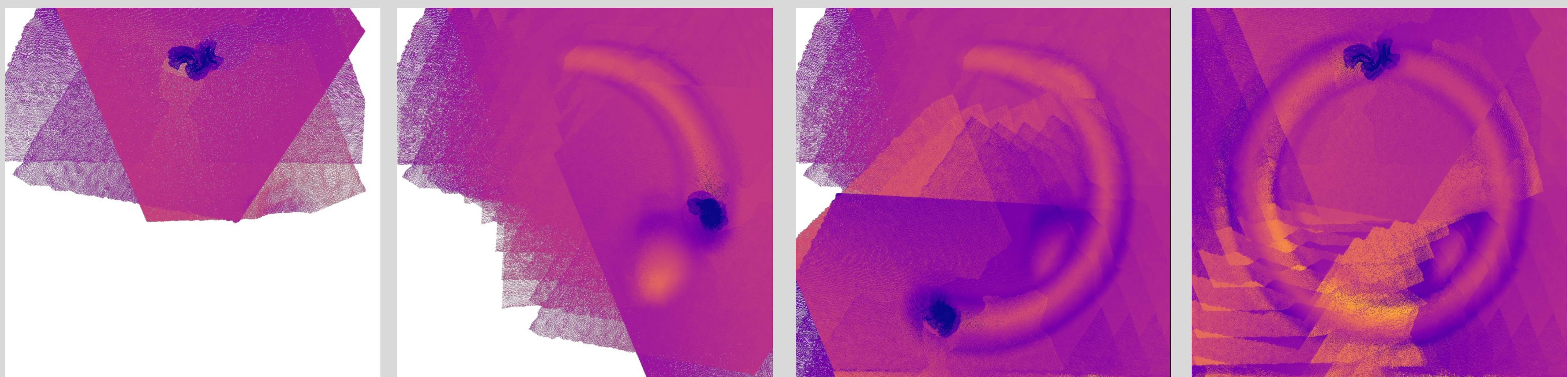
- Heightmaps are produced by vertically binning points into a grid-based format
- Turns a point cloud into an easily computable matrix
- Each cell encodes height information
- Lowers computation and memory storage requirements



An indentation transformed into a heightmap

Unfiltered multiple camera results

- Simultaneous displacement and mapping
- White space represents unknown terrain
- As the tool traverses and deforms the terrain, the circular deformation can be clearly seen



Dynamic surface mapping

Future work

- Fine tune point cloud alignment in real-time using an Iterative Closest Point (ICP) method
- Collect datasets which contain the force-displacement history, using a six-axis load cell
- Assess methodologies for automated earthmoving

1: E , D , and ${}^{\text{tool}}T_{\text{cam}}$ are 4x4 homogenous transformation matrices and points p_i are 4x1 column matrices



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